meVBT

VELOCITY-BASED TRAINING · MUSCULAR RECOVERY

INVESTOR BRIEF · JUNE 2026

Your heart-rate wearable can't see the barbell.
We give your muscles their own recovery score.

Whoop and Apple Watch read a near-failure squat session as a rest day — HR barely moves while your muscles grind to zero. **meVBT counts every rep and measures intra-set velocity loss** — the validated proximity-to-failure signal — from the phone and watch you already own. No bar sensor. No setup. No taps.

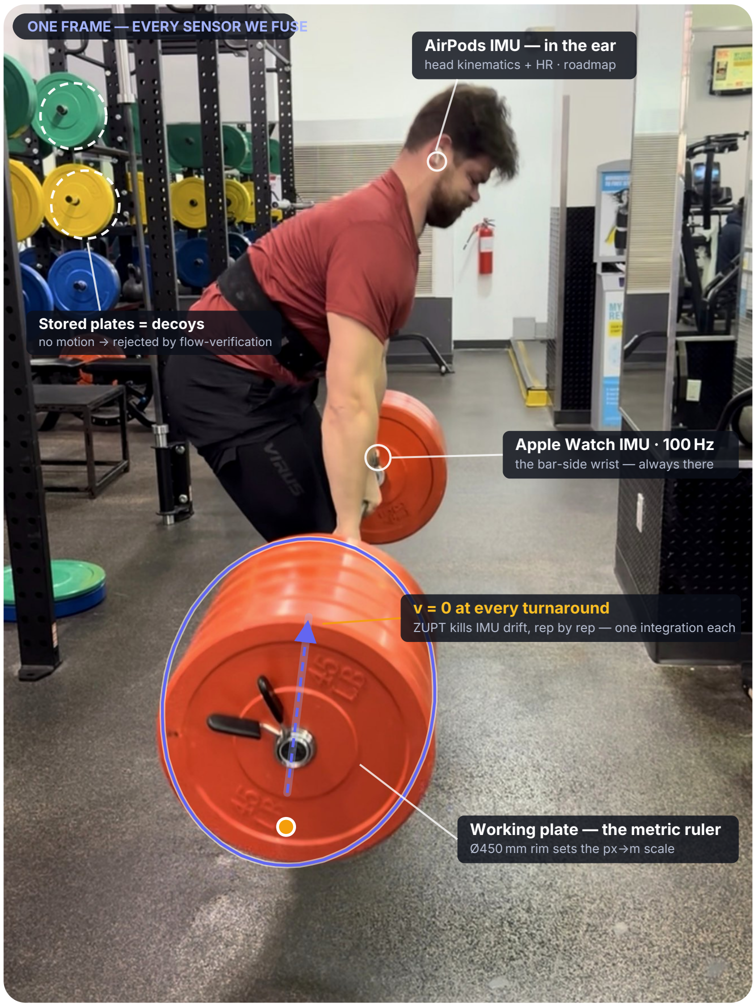
8×
more accurate rep counts than the leading video app — 0.32 vs 2.57 mean error, head-to-head on 22 real sets

3×
better fatigue signal — velocity-loss within 3.2pp of lab hardware; the incumbent: 9.0pp

21/22
sets within ±1 rep, fully automatic — every bench & squat set exact

$0
added hardware — replaces a $500 bar-mounted transducer with software

01 · THE SIGNAL, ON ONE REAL FRAME



Every signal we fuse, in one real frame. The watch rides the bar-side wrist; the AirPods add a second IMU; the plate's Ø450 mm rim hands video its metric scale; the stored plates behind are the decoys the tracker must reject. Overlays here are the sensor anatomy — p. 2 shows the tracker's raw output.

WHAT EACH SENSOR KNOWS

Phone camera — plate CV — SHIPPING SIGNAL

Zero-tap rep counting + velocity-loss that already beats the market leader (left, p. 2). Works on bumpers, dark iron, hex plates, mirrors.

Apple Watch IMU — POC BUILT

100 Hz gravity-corrected acceleration → ZUPT-anchored velocity. Always on your wrist — the only sensor that's *always there*.

BLE bar devices & LPT — CALIBRATION GT

Vitruve linear transducer = our ground truth; a 6-vendor dataset quantifies every tool's real error. Ingested, never required.

AirPods IMU + HR — ROADMAP

Head kinematics for the lifts the wrist can't see; HR closes the systemic-vs-muscular loop.

→ One fusion model

Every source emits the same shape — rep boundaries, velocity profile, ROM, **per-rep confidence**. A learned personal prior (your rep shape, your tempo, your minimum velocity) strengthens with every set; every manual correction trains it further.

Muscular Strain & Recovery Score

the lifting analogue of Whoop — structurally impossible from heart rate alone

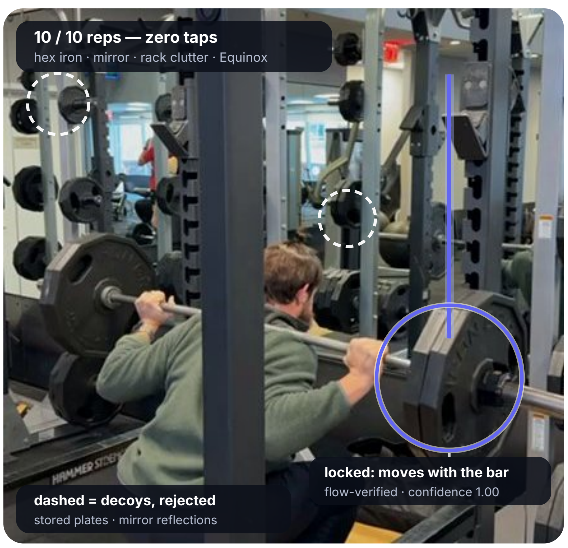
02 · THE PROOF

We already beat the market's best lifting CV — and we're just getting started.

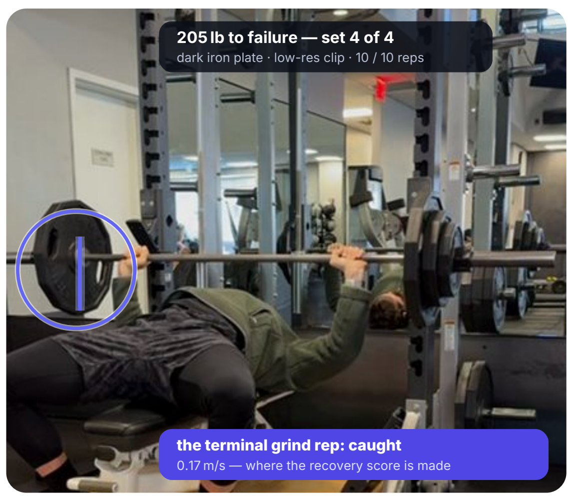
26 real training clips, four gyms, 2024–2026 — bumper, dark-iron and hex plates; side, diagonal and front angles; mirrors, racks, occlusion. Ground truth: a Vitruve bar-mounted linear transducer. SmartBarbell (the leading velocity-video app) scored on identical clips. Our system gets **only the raw clip** — no taps, no hints, no gym profile.

	meVBT CV fully automatic · phone only	**SmartBarbell** leading video app	**Vitruve LPT** $500 hardware = ground truth
COUNTING EVERY REP			
Mean rep-count error per set, vs ground truth	**0.32** 8× better	2.57	reference
Sets counted exactly	**16 / 22**	7 / 21	—
Sets within ±1 rep	**21 / 22**	13 / 21	—
Main lifts (every bench & squat set)	**100% exact**	misses up to 7 reps/set	—
THE FATIGUE SIGNAL — INTRA-SET VELOCITY LOSS			
Velocity-loss error pp vs ground truth, same 11 clips	**3.2 pp** ~3× better	9.0 pp	reference
Worst-case fatigue read squat set, true loss 30%	**26%**	2.4% — reads near-failure as fresh	30%
TRUST & EXPERIENCE			
Knows when it can't measure	**abstains + flags, per rep**	reports phantoms at 100% confidence	n/a
Setup per set	**none — point the phone**	aim + confirm	mount unit, clip to bar
Hardware cost	**$0**	$0	~$500

Benchmark frozen 2026-06-11; every number regenerates from the repo (cv_eval.py / vel_eval.py). Ground truth is scoring-only — never an input. SmartBarbell n=21 (no reading on one clip).



The hard part is knowing what to track. We propose every moving circle, flow-verify each, and keep the one that locks with plausible, regular reps. Method, not gym profile.



Near-failure reps are the product. They carry the fatigue signal — and they're exactly where the incumbent loses the bar (below).

VELOCITY-LOSS READ
same bench set →

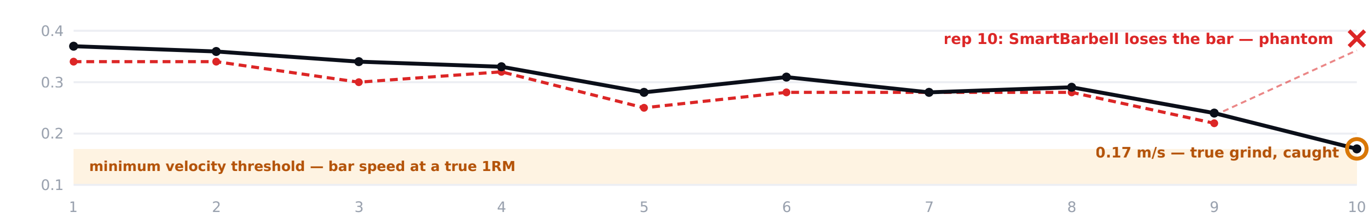
46.0%
meVBT — error 1.4 pp

44.6%
Vitruve ground truth

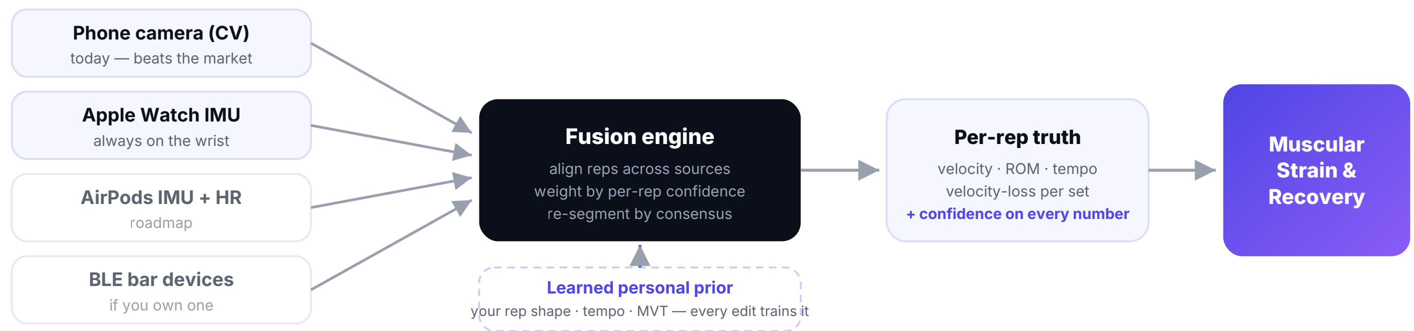
missed
SmartBarbell — lost the terminal rep

One phantom or missed rep at the grindy finish corrupts the whole fatigue read — counting and fatigue are one problem.

The rep that matters most — per-rep bar velocity, bench 205 lb × 10 to failure black = Vitruve ground truth · red = SmartBarbell



Computer vision is the wedge.
Fusion + the recovery score is the moat.



Confidence is open white space

No competitor surfaces measurement confidence — we watched one report a phantom rep at a 100% badge. Per-rep, per-metric confidence is engineered in from day one, and it's what makes fusion possible.

HR platforms structurally can't follow

Whoop's strain is cardiovascular by construction. The muscular recovery score requires counting reps and reading bar-speed decay — a sensing-and-fusion problem, not a heart-rate feature.

Device-agnostic by design

Every source emits the same interface. The watch is the most convenient sensor, not a dependency — the same model ingests Garmin, Fitbit, a phone on a bench, or nothing but video.

The lab — a measurement discipline competitors don't have

34	2,720	6	26	8	4
MULTI-VENDOR SETS	REP-LEVEL MEASUREMENTS	VENDORS BENCHMARKED	SCORED VIDEO CLIPS	LIFTS COVERED	GYMS · ALL ANGLES

Every commercial tool recorded side-by-side on the same physical sets, aligned rep-by-rep. This is how we know the market's real error bars — and it becomes the seed prior for every new user's model. No-cheating evals: the system sees only the clip; ground truth is scoring-only; every number regenerates from the repo.

THE PATH — EACH PHASE SHIPS STANDALONE VALUE

PHASE 0 DONE
Capture lab
watch→phone pipeline · multi-vendor dataset

PHASE 1 WON
CV wedge
beat the best video app on counts + fatigue

PHASE 2
Calibrate watch
watch vs Vitruve; quantified per-lift error

PHASE 3
Watch-only loop
confidence + personal prior, on the wrist

PHASE 4
Fuse
video + watch + editor → consensus reps

PHASE 5
The score
muscular strain & recovery, any lift

Velocity zones (mean concentric, m/s)

strength 0–.5 | .5–.75 | .75–1.0 | speed 1.0+

We prefer personal rolling baselines over fixed zones — your bar speed, not a textbook's.

Velocity-loss cut-offs

`~10%` power · `~20%` strength · `~30%+` hypertrophy. Validated proximity-to-failure — the basis of the strain score.

Minimum velocity threshold

Bar speed at true 1RM: **`deadlift .15–.20`** · **`bench .10–.17`** · **`squat .25–.30`**. Personalized per lifter in our priors.

WHAT WE'LL SAY IS NOT SOLVED YET

Absolute m/s from low-res video trails the incumbent (~0.07 m/s) — so we abstain and report the scale-free fatigue signal rather than guess; HD capture and a learned plate-sizer close it. Dead-front camera angles defeat every tool (including the incumbent). Single-dumbbell accessories await the pose + watch path. We publish our limits — it's the same honesty the confidence system sells.